describe the behaviour of the electric and the magnetic fields: the mathematical version of the ‘Faraday lines’.[fn9]

Today, Maxwell’s equations are used daily to describe all electric and magnetic phenomena, to design antennae, radios, electric engines and computers. And this is not all: these same equations are needed to explain how atoms function (they are held together by electrical forces), and why the particles of the material that forms a stone adhere together, or how the Sun works. They describe an amazing number and range of phenomena. Almost everything that we witness taking place – with the exception of gravity, but little else besides – is well described by Maxwell’s equations.

But there is more. There is still what is perhaps the most beautiful success of science: Maxwell’s equations tell us what light is.

Maxwell realizes that his equations predict that Faraday’s lines can tremble and undulate, just like the waves of the sea. He computes the speed at which the undulations of Faraday’s lines move and the result turns out to be … the same as for light! Why? Maxwell understands: because light is nothing other than this rapid trembling of Faraday’s lines! Not only have Faraday and Maxwell figured out how electricity and magnetism work but, with the same stroke, as a collateral effect, they have figured out what light is.

We see the world around us in colour. What is colour? Put simply, it is the frequency (the speed of oscillation) of the electromagnetic wave that light is. If the wave vibrates more rapidly, the light is bluer. If it vibrates a little more slowly, the light is redder. Colour, as we perceive it, is the psychophysical reaction of the nerve signal generated by the receptors of our eyes, which distinguish electromagnetic waves of different frequencies.

I wonder how Maxwell felt when he realized that his equations – written to describe bobbins, small cages and little needles in Faraday’s lab – turned out to explain the nature of light and colour …

Light is thus nothing more than a rapid vibration of the spiderweb of Faraday’s lines, which ripple like the surface of a lake as the wind blows. It isn’t true that we ‘do not see’ Faraday lines. We *only* see vibrating Faraday lines. ‘To see’ is to perceive light, and light is the movement of Faraday lines. Nothing leaps from one location in space to another without something transporting it. If we see a child playing on the beach, it is only because between them and ourselves there is this lake of vibrating lines which transport their image to us. Is the world not marvellous?

The discovery is extraordinary, but there is even more. The ultimate corollary of the discovery has a concrete value for us that is unequalled. Maxwell recognizes that the equations foresee that Faraday’s lines can also vibrate at much lower frequencies, that is to say, more slowly than light. Therefore, there must be *other* waves which nobody had yet seen, produced by the movement of electrical charges, which in turn move electrical charges.